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Mini-Project
REPORT

TRUE-RMS-TO-DC CONVERTER

PROJECT TEAM ID

Group A2

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1. Introduction

The root mean square (RMS) value of a time-varying voltage is the equivalent DC voltage that would dissipate the same average power in a resistive load [1]. For a signal *V(t)* observed over a period *T*, it is defined as

$$Vrms = \sqrt{\frac{1}{T}\int_0^T V^2(t)\,dt} \tag{1.1}$$

A true RMS-to-DC converter evaluates Eq. (1.1) continuously and produces a steady DC output equal to the RMS value of any input waveform, including composite AC+DC signals. However, many low-cost meters instead measure the rectified average and scale it by a constant chosen for sine waves, so they are accurate only for sinusoidal inputs. Their errors become severe for pulse-like signals with a high crest factor, the ratio of peak to RMS value [1]. Real signals are rarely pure sinusoids, so accurate measurement requires a converter that computes the RMS value directly. True RMS conversion is therefore essential wherever signal power must be measured reliably. Applications include power-quality monitoring of non-sinusoidal mains loads, audio level metering, test and measurement instrumentation such as true RMS multimeters, and industrial control systems [2], [3]. This project designs, simulates, and builds a fully analogue true RMS-to-DC converter for the ECE3161 mini project [3]. Under the project constraints, no digital devices may be used and only components from the supplied kit are permitted. The circuit is powered from the ±5 V rails of the Analog Discovery 2/3. It must accept inputs of 100–200 mVpp with DC offsets from −3 V to +3 V, frequencies from DC up to 20 kHz, and crest factors from 1 to 3. Its input impedance must be at least 1 MΩ with an input capacitance of no more than 50 pF, its output error must stay below 10%, and it must settle within 10 s. Because the input carries a DC offset, the converter must report the combined (AC+DC) RMS value; for the specified calibration input, this is 1.0025 V [3]. To meet these requirements, an implicit log–antilog converter was designed using discrete op-amps and transistors from the kit. Section 2 surveys RMS conversion techniques and justifies this choice, Sections 3 to 6 present the specifications, design, implementation, and results, and Section 7 concludes.

2. Literature Survey

This section reviews the main approaches to measuring the RMS value of a signal, from average-responding meters to true RMS converters, and assesses each against the project requirements. It then examines how commercial converters implement the selected approach and how each block of the proposed design draws on that literature.

2.1 Average-Responding (MAD) Measurement

The most common AC measurement is not a true RMS method. A precision rectifier and filter compute the mean absolute value (MAD) of the signal, which is then multiplied by 1.11, the RMS-to-MAD ratio of a sine wave [1]. Because this ratio is 1.00 for a square wave and 1.155 for a triangle wave, a sine-calibrated meter reads them with errors of +11% and −3.8% respectively. The error grows rapidly as the duty cycle of a pulse train falls, while a true RMS converter remains considerably more accurate [1]. This method is therefore unsuitable, although its precision rectifier is reused in Block 1.

2.2 True RMS Conversion Methods

To determine the RMS value of an arbitrary waveform, true RMS converters evaluate Eq. (1.1) directly instead of assuming a sine wave. They fall into three groups: thermal converters, which measure the heating effect of the signal, computing converters, which square, average, and square-root it electronically, either explicitly or implicitly, and more recent delta-sigma or digital converters. Each group is outlined below and assessed against the project requirements in Section 2.3.

Thermal conversion compares the heat produced by the input in one resistor with that produced by an adjustable DC reference in a matched resistor, and uses feedback to adjust the reference until the two temperatures are equal [1]. It achieves about 0.1% error and wide bandwidth, but it is the most difficult and expensive method, its thermal time constant limits low-frequency performance [1], and it requires matched heater–sensor pairs that the kit does not contain.

Computing methods square, average, and square-root the signal electronically. *Explicit computation* performs these operations in sequence and can reach ±0.1% error. However, its squarer output spans the square of the

input range, so a 100:1 input produces a 10,000:1 internal range, which restricts practical designs to about 10:1 [1]. *Implicit computation* instead divides the squared input by the output before averaging. Since the output is a slowly varying DC level, it can be taken outside the average [1]:

$$V_{out} = \left(\frac{\overline{V_{in}^2}}{V_{out}}\right) \Rightarrow V_{out}^2 = \overline{V_{in}^2} \quad (2.1)$$

The feedback therefore performs the square root. Because the internal signal V_{in}^2/V_{out} scales linearly with input level, dynamic range is greatly increased, at the cost of lower bandwidth [1]. The squaring and division are commonly implemented with log–antilog circuits, as in the AD536A, AD636, and AD637 [1], [4], [5].

Delta-sigma and digital methods, such as the LTC1966 [6] or ADC-based computation, offer high accuracy but rely on switched or digital circuitry, which the project forbids [3].

2.3 Method Selection Against Project Requirements

The smallest specified input [3], a 100 mVpp triangle wave, has an RMS value of 28.9 mV. The largest, a 200 mVpp sine on a 3 V offset, has an RMS value of 3.0 V. Together they span a range of roughly 100:1. An explicit squarer would need about 10,000:1; within ±5 V rails, its smallest outputs would fall below a millivolt, comparable to the input offsets of the kit's op-amps. Average-responding, thermal, and digital methods fail on accuracy, buildability, and the analogue-only rule respectively. Only implicit log–antilog computation meets every requirement and can be built from the kit's op-amps and bipolar transistors, and its reduced bandwidth is acceptable at 20 kHz. This approach was therefore adopted.

2.4 Monolithic Implementations and Design Linkage

The AD536A rectifies the input, converts it to a current, and computes $I_4 = I_1^2/I_3$ with a four-transistor translinear squarer/divider. It averages in current mode with a fixed internal resistor [1], so its settling time lengthens at low input levels [1]. The AD636 is a low-power variant [1]. The AD637 instead averages with a separate op-amp filter, giving settling time independent of input level, and laser trimming gives it higher accuracy and bandwidth [1], [4]. The proposed design combines the AD536A's squarer/divider with AD637-style averaging. Because off-kit ICs are forbidden [3], it replaces laser trimming and on-die matching with manual trims and a calibrated gain adjustment.

3. Design Specifications

3.1 Requirements Analysis

The formal project requirements and their implications for circuit design and verification are summarised in **Table 3.1**.

Table 3.1: Requirements analysis and verification plan

Criterion	Formal requirement [3]	Design implication	Verification
Power supply	±5 V rails; peak current ≤ 200 mA	Use rail-compatible op-amps, local decoupling and internal signal levels that avoid saturation.	Measure total supply current and inspect internal node limits.
Input magnitude and offset	100 to 200 mVpp; −3V to +3V DC offset	Use a precision rectifier because the AC signal is below a normal junction drop; preserve both AC and DC contributions.	Test both amplitude limits and offsets at −3 V, 0 V and +3 V.
Waveform and crest factor	DC, sine, square, triangle, rectified and complex waveforms; CF = 1 to 3	Use true-RMS computation rather than sine-calibrated average rectification; prevent internal saturation at high peaks.	Test representative waveforms, including CF = 1, 2 and 3 cases.
Frequency response	DC to 20 kHz	Select suitable op-amps and minimise parasitic capacitance; rectifier switching and slew rate are critical at high frequency.	Verify DC, 1 kHz, 10 kHz and 20 kHz operation.

Input loading	Rin ≥ 1 MΩ; Cin ≤ 50 pF	Use the 1 MΩ TL084 input stage and short input wiring to minimise source loading.	Confirm resistance from component values; estimate or measure input capacitance.
Accuracy	Formal output error < 10%	Provide offset, rectifier-balance and gain trims to compensate for op-amp offset, resistor tolerance and BJT mismatch.	Compare theoretical, LTspice and measured RMS values after one fixed calibration.
Output and settling	Rout ≤ 1 MΩ; settling time ≤ 10 s	Use an averaging network followed by a low-impedance unity-gain buffer; select the time constant as a ripple–response compromise.	Measure a step response using a stated tolerance band and verify output loading.

3.2 Design Goals

The principal design goal is to maintain accurate RMS conversion across the complete specified input range. Agreement within 1% among theoretical, simulated and practical results is targeted. The converter is also intended to maintain accurate, low-distortion operation through the 20 kHz limit. Settling should occur within 3s without oscillation or significant drift and should remain consistent for low-level, DC-offset and high-crest-factor inputs. These goals are verified after a single calibration, with the same trimmer settings retained for all test conditions.

4. Design Description

The proposed True RMS-to-DC converter uses an analogue log–antilog circuit topology, which calculates the RMS value of input voltages through nonlinear transistor characteristics and negative feedback. The converter consists of six functional blocks to perform the equivalent mathematical operations of squaring, averaging and square-root of different waveforms. The design was then implemented in LTSpice, where each block was verified before calculating the final RMS value under different waveforms. The main components include TL084 and TL974 op-amps, 2N3904 BJTs, resistors and capacitors.

4.1 Theoretical Design

The design is based on the RMS definition in equation (4.1), where T represents the averaging period.

$$VRMS = \sqrt{\frac{1}{T}\int_0^T Vin^2(t)\,dt} = \sqrt{average(V^2(t))} \tag{4.1}$$

Block 0: Inverting Input Buffer

The first block uses an inverting TL084 op-amp that effectively acts as a buffer to isolate the input signal source from the loading effects of the subsequent rectification stage through its near zero output impedance. The output is as expressed in equation (4.2).

$$V1 = -\frac{Rf1}{Rin1}Vin = -\frac{1\,M\Omega}{1\,M\Omega}Vin = -Vin \tag{4.2}$$

With Rf1 = Rin1 = 1 MΩ, the ideal closed-loop gain is -1, producing an inverted signal with the same amplitude. The 1 MΩ input resistor achieves the input resistance required by the project specification, and the JFET-input TL084 is selected to minimise input bias-current effects when operating with large value resistors.

Block 1: Precision Full-Wave Rectifier

At the first stage, when v1 > 0, QD2 conducts while QD1 is off, allowing U2 to operate as an inverting amplifier with gain of -1, where va = -v1. Conversely, when v1 < 0, QD1 conducts while QD2 is off, causing va to remain approximately zero. Thus, these transistors control the conduction paths for opposite input polarities. Unlike a conventional diode rectifier, the op-amp feedback compensates for the transistor junction voltage drop, allowing relatively small input signals (100 - 200 mVpp) to be rectified.

At the second stage, U3 acts as an inverting summing amplifier that combines v1 and va:

$$vr = -\left(\frac{Rf3}{R3a}v1 + \frac{Rf3}{R3b}va\right) = -\left(\frac{20k}{20k}v1 + \frac{20k}{10k}va\right) = -v1 - 2va \tag{4.3}$$

For a positive input, va = -v1, giving vr = v1, for a negative input, va = 0, giving vr = -v1. Hence, both positive and negative input polarities produce a positive output.

Block 2: Logarithmic Squarer

The purpose of Block 2 is to convert the full-wave rectified voltage, vr, into a logarithmic voltage representing the squared input signal. The operation is based on the exponential collector current relationship of a BJT, where the two diode-connected transistors Q1 and Q2 each produce a logarithmic base-emitter voltage in equation (4.4). I_s represents the saturation current, n is the emission coefficient, and V_T is the thermal voltage, approximately 25.7 mV at 25℃. Since the non-inverting input of U4 is grounded, the input current through Rsl gives $I_{in} = vr/R_{sl}$.

$$Vsl = -(V_{BE1} + V_{BE2}) = -2nV_T ln(\frac{I_{in}}{I_s}) \tag{4.4}$$

Using $2\ln(x) = \ln(x^2)$, the output represents the square of the input current in logarithmic form. The TL084 is selected for its high input impedance and low input bias current, which reduce errors in logarithmic conversion. QD3 diode acts as a safety clamp that restores negative feedback if the logarithmic junctions stop conducting, preventing op-amp output from saturating at the power supply rails. The 1 kΩ emitter resistor RE4 limits current, while the 22 pF compensation capacitor Cc4 improves high-frequency stability.

Block 3: Feedback Logarithmic

This block receives the averaged output voltage vl from Block 4 through R5 and converts it into a logarithmic feedback voltage vfl. Since U5 non-inverting input is grounded, the feedback current is $I_{fb} = vl/R5$. Using BJT,

$$Vfl = -nV_T ln(\frac{I_{fb}}{I_s}) \tag{4.5}$$

This feedback voltage is applied to the base of Q4, allowing the circuit to regulate Q4's collector current according to the output voltage. QD5 diode, Cc5 capacitor and RE5 emitter resistor have the same function as mentioned in Block 2.

Block 4: Averaging and Square-Root Conversion

This block functions to average the varying Q4 collector current and produce a constant DC output voltage vl. The logarithmic outputs from Blocks 2 and 3 are applied to the emitter and base of Q4 respectively, giving $V_{BE4} = vfl - vsl$. Using the BJT exponential relationship and substituting equation (4.4) and (4.5),

$$I_{Q4} = I_s e^{(vfl-vsl)/nV_T} = K\frac{I_{in}^2}{I_{fb}} = \frac{v1}{R7} + C7\frac{dvl}{dt} \tag{4.6}$$

where K represents the effective transistor-matching factor. Thus, Q4 creates negative feedback. When Q4 current exceeds the current through R7, C7 charges and vl increases. Conversely, when Q4 current is lower, C7 discharges and vl decreases. The output voltage vl is continuously fed back to Block 3, which adjusts vfl and consequently Q4 current until its average balances the current through R7. At steady state, the capacitor current approaches zero, by combining with equation (4.6), assuming vl varies slowly relative to the input and K = 1,

$$vl^2 = K\frac{R7\,R5}{Rsl^2}\overline{vr^2} = K\frac{(10k)(10k)}{(10k)^2}\overline{vr^2} \;\; where\; vl = \sqrt{\overline{vr^2}} = V_{RMS} \tag{4.7}$$

Thus, the output naturally settles when the average Q4 current balances the current through R7, achieving the implicit square-root operation. The 10 µF capacitor C7 smooths rapid current variations and performs averaging, where the time constant is calculated to be $\tau = R7C7 = (10k\Omega)(10\mu F) = 0.1s$.

Block 5: DC Scaling and Output Buffer

The resistor Rp and capacitor Cp form a first order RC low-pass filter, which attenuates the remaining high-frequency AC ripple while preserving the DC component, vl from Block 4 representing the RMS voltage. The cutoff frequency is calculated as 1.59 Hz, providing a compromise between ripple reduction and settling time.

$$f_C = \frac{1}{2\pi RpCp} = \frac{1}{2\pi(10k\Omega)(10\mu F)} = 1.59\,Hz \tag{4.8}$$

U7 is configured as a unity gain buffer, where Vout = filtered vl. This buffer provides high input impedance to prevent loading of the RC filter, while its low output impedance enables accurate measurements without significantly affecting output voltage.

4.2 Practical Considerations

The practical implementation uses components available in the ECE3161 laboratory kit as shown in **Table 4.2.1**. Since some theoretical components are unavailable, suitable resistor combinations are required.

Table 4.2.1: Mapping of theoretical components to practical implementation

Component	LTspice design	Practical Implementation
Op-amps	TL084, TL974	TL084CN, TL974IN
Transistors (Diode)	2N3904	Base-collector connected 2N3904
Resistors	1 MΩ, 10kΩ, 20kΩ, 1kΩ	Kit 1 % metal-film resistors; 20 kΩ = 2 × 10 kΩ in series
Capacitors	22 pF, 100 nF, 10µF	Kit ceramic (22 pF, 100 nF) and 10 µF electrolytic capacitors

Noise Impact: Practical transistors exhibit variations in saturation current, base-emitter voltage and temperature-dependent characteristics, which may introduce errors in the logarithmic conversion stages. Similarly, op-amp input offsets, bias currents, resistor thermal noise and component tolerances can affect RMS accuracy, where small current variations can affect the transistor base-emitter voltages, particularly when processing small input signals of 100–200 mVpp. To minimise these effects, the TL084 was selected for its low input bias current. Additionally, the 22 pF compensation capacitors improve the stability of the logarithmic amplifiers, whereas the 10 µF averaging and output-filter capacitors reduce high-frequency fluctuations. Additionally, 100 nF supply decoupling capacitors and 10 µF bulk bypass capacitors are added to help minimise supply noise and external interference. Since these measures cannot completely eliminate component mismatch and offset errors, potentiometers were introduced in the physical implementation for gain and offset calibration.

4.3 Practical Design and Simulation

The True RMS-to-DC converter was implemented and analysed using LTspice as shown in Figure 4.3.1.

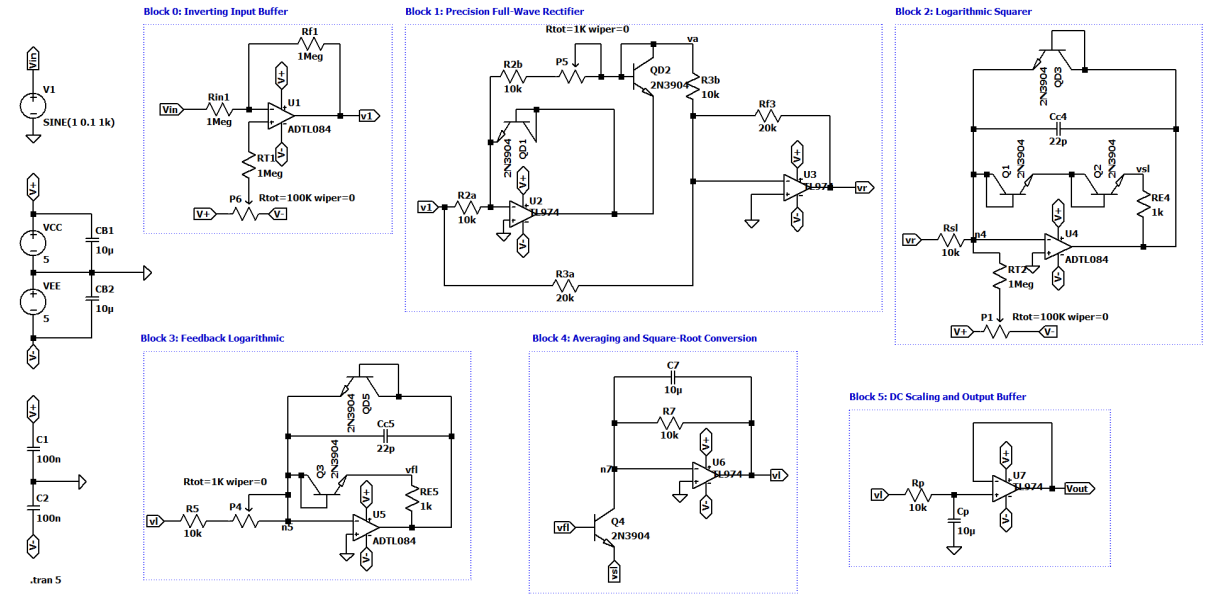
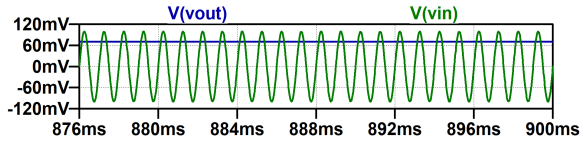


Figure 4.3.1: Schematic diagram of proposed theoretical design in LTspice

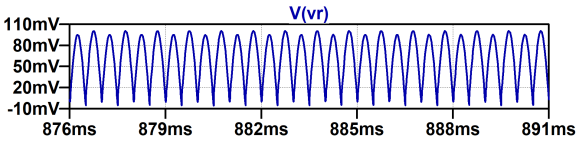
Transient analysis was performed to evaluate the circuit performance under different input conditions. The four potentiometers are added for further calibration in practical implementation as discussed in Part 5 later, so the wiper is set to 0 in LTSpice simulations. The data collected are summarized in **Table 4.3.1**, where the circuit design was tested against twelve test cases, and the simulated output voltage was measured at Vout.

Table 4.3.1: RMS Measurement Results in LTSpice Simulation

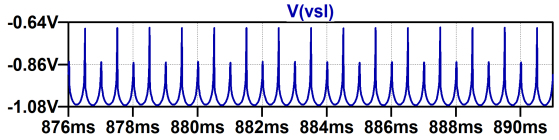
Test Case	Waveform Type	V_{pp} / Parameter	DC Offset	Frequency	Simulated RMS Value	Theoretical RMS Value	% Error
CAL	Sine	200 mV_{pp}	+1 V	1 kHz	1.0026	1.0025	+0.01
1	DC	—	+2 V	DC	2.006	2.000	+0.30
2	DC	—	−3 V	DC	3.002	3.000	+0.07
3	Sine	200 mV_{pp}	0 V	1 kHz	0.07080	0.07071	+0.13
4	Sine	200 mV_{pp}	0 V	10 kHz	0.07013	0.07071	-0.82
5	Sine	100 mV_{pp}	0 V	20 kHz	0.03539	0.03536	+0.08
6	Triangle	100 mV_{pp}	0 V	1 kHz	0.02880	0.02887	-0.24
7	Square	200 mV_{pp}	0 V	1 kHz	0.0992	0.1000	-0.78
8	Full-wave rect. sine	0–200 mV	0 V	1 kHz	0.1415	0.1414	+0.07
9	Half-wave rect. sine	0–200 mV	0 V	1 kHz	0.1007	0.1000	+0.70
10	PWM, 11.1 % duty	200 mV_{pp} (0–200 mV)	+100 mV	1 kHz	0.06654	0.06667	-0.19
11	Sine	200 mV_{pp}	−2 V	5 kHz	2.0028	2.0013	+0.07
12	Square	100 mV_{pp}	+3 V	10 kHz	3.006	3.0004	+0.19



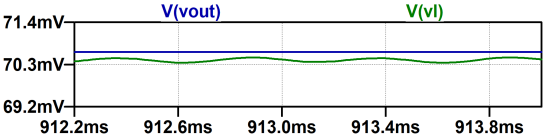
(a) Sine Wave Input as Green and DC output as Blue



(b) Full-wave Rectifier Output



(c) Square and Logger Output



(D) Feedback Logger Output as Green and DC RMS as Blue

Figure 4.3.2: LTspice simulation results for a 200 mVpp, 1 kHz sine-wave input with zero DC offset: (a) input and DC output, (b) full-wave rectified signal, (c) logarithmic squarer output, and (d) averaged RMS and final DC output.

Overall Accuracy: The LTspice simulation results show good agreement with the theoretical RMS values. All twelve test cases have errors within ±0.82%, meeting the project's 1% simulation accuracy target. The calibration case produces 1.0026 V against the expected 1.0025 V, giving an error of +0.01%. This demonstrates that the fixed-component design achieves high accuracy without additional potentiometer adjustments.

Frequency Response: At 200 mVpp, the sine-wave error changes from +0.13% at 1 kHz to −0.82% at 10 kHz, potentially due to the finite bandwidth of the op-amps and transistor switching effects, but still demonstrate satisfactory frequency performance for the tested conditions.

Waveform and Crest Factor: Different waveform shapes were tested to verify true RMS operation. The triangle, full-wave rectified sine and half-wave rectified sine produce errors of −0.24%, +0.07% and +0.70%, respectively, while the PWM waveform with a crest factor of 3 gives an error of −0.19%. This indicates that the converter accurately measures RMS values across different waveform shapes.

DC and Offset: The +2 V and −3 V DC inputs produce simulated outputs of 2.006 V and 3.002 V, corresponding to errors of +0.30% and +0.07%, respectively. This demonstrates the rectifier's ability to handle both input polarities while preserving the RMS magnitude.

Settling Time and Design Limitations: Other design limitations include the finite bandwidth of operational amplifiers and the trade-off between output ripple and settling time. At higher input frequencies, the limited response speed of the op-amps may introduce rectification errors and reduce RMS accuracy. Meanwhile, increasing the averaging capacitance improves output smoothing but also increases the settling time. Therefore, C7 and Cp were selected as 10 µF to balance ripple reduction and response speed, and the settling time is safely constrained within 1 second. In addition, op-amp offsets and BJT temperature parameter variations may affect logarithmic conversion, particularly at low input amplitudes. The 22 pF compensation capacitors improve feedback stability, while supply decoupling and short wiring help reduce noise and interference.

5. Physical Implementation

The converter was built on two joined breadboards using only parts from the unit kit, powered from the AD3 ±5 V supply (*Fig. 5.1*): +5 V and GND run along the top rails, and −5 V and GND along the bottom rails. Signal flows from right to left. The right-hand board holds the input buffer and rectifier (Blocks 0–1), and the left-hand board holds the translinear core, averager and output buffer (Blocks 2–5). A jumper lead at each block output lets every node be probed in turn during testing and the demonstration.

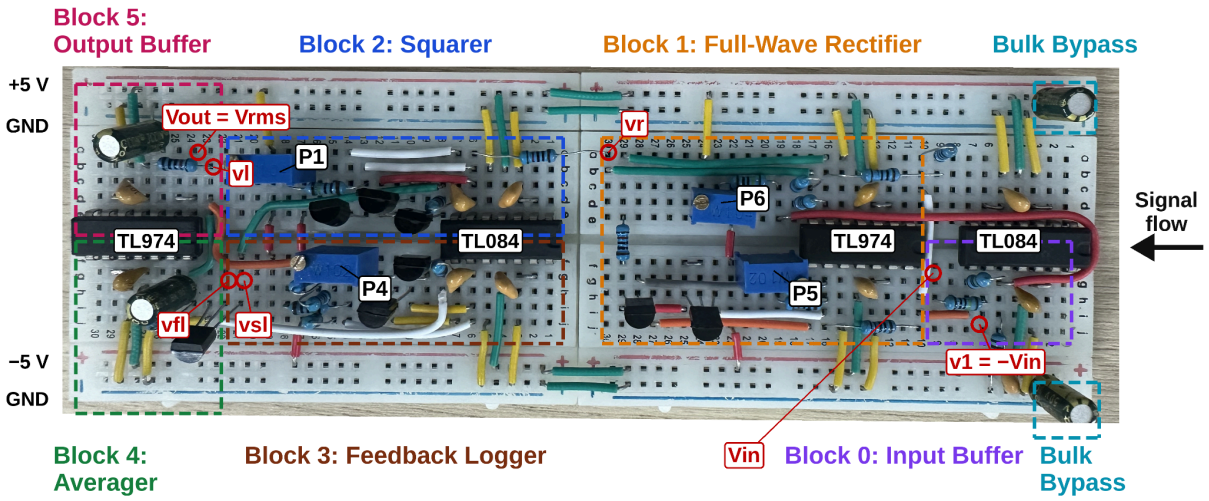


Figure 5.1: Breadboard implementation of the True RMS-to-DC converter. Dashed boxes mark Blocks 0–5 and the bulk bypass capacitors, and each op-amp IC is labelled TL084 or TL974; red tags mark the input node (driven by AD3 W1 and probed on C1) and the probe point at each block output; black tags mark the trimmers.

5.1 Component Choices and Practical Modifications

Table 5.1 maps each block to the parts used and the practical changes made from the theoretical design in Section 4.

Table 5.1: Block implementation and practical modifications

Block	Function	Parts used	Practical consideration
0 - Buffer	Inverting input buffer	TL084 (U1), 1 % resistors, P6	P6 trims op-amp input offset; the 1 MΩ input resistor meets the ≥ 1 MΩ input-impedance spec
1 - Rectifier	Precision full-wave rectifier	TL974 (U2), 2N3904 × 2 (T7, T8, diode-connected), 1 % resistors, P5	P5 matches the negative-input gain to the positive-input gain
2 - Squarer	Logs $\|V_{in}\|$ across two V_{BE} drops	TL084 (U3), 2N3904 × 3 (T1–T3), P1	P1 trims the residual offset after the rectifier
3 - Feedback logger	Logs V_{out} to close the translinear loop	TL084 (U3), 2N3904 × 2 (T4, T5), P4	P4 sets overall gain, absorbing V_{BE} mismatch between unmatched 2N3904s

4 - Averager	Averages the squarer output	TL974 (U4), 2N3904 (T6), 10 µF electrolytic	Only usable electrolytic in the kit; sets the settling time (Section 6.2)
5 - Output	Buffers the DC output to the meter	TL974 (U4), 10 µF electrolytic	Low output impedance, well below the 1 MΩ limit

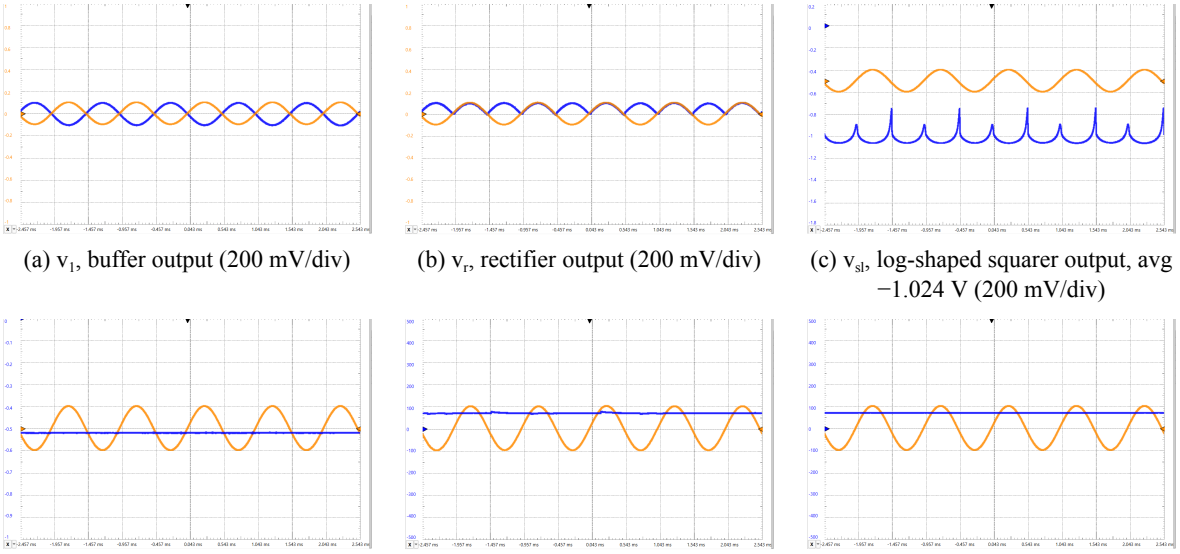
Four practical measures kept the hardware close to the design. First, four multi-turn trimmer potentiometers (P1, P4, P5, P6) were built in so that component tolerances and op-amp offsets could be tuned out on the board: P6 and P1 null the offsets, P5 balances the negative-input gain, and P4 sets the overall gain so that the output matches the calibration case (1.0025 V). Second, the translinear transistors are unmatched discrete 2N3904s, so their V_{BE} mismatch is absorbed by the P4 gain trim rather than by device selection, and they were placed side by side so that they share the same temperature, which limits drift. Third, each op-amp supply pin was decoupled with a 100 nF ceramic capacitor.

5.2 Sub-Block Verification

Block outputs for CAL and case 3 (*Table A.1, Appendix A*) confirm that each stage behaves as designed (*Figure 5.2*). The feedback logger follows the expected logarithmic law: between CAL (V_{out} = 1.0038 V) and case 3 (V_{out} = 70.87 mV) its voltage changed by 70.9 mV, which gives Eq. (5.1).

$$nV_T = \frac{\Delta V_{fl}}{ln(V_{out,CAL}/V_{out,3})} = \frac{70.9mV}{ln(1.0038/0.07087)} = 26.7mV \quad (5.1)$$

This matches V_T = 25.7 mV at 25 °C with an emission coefficient n ≈ 1.04 for the 2N3904. Because nV_T appears at every junction of the translinear loop, it cancels in the output. V_{BE} mismatch between the discrete transistors does not cancel: at CAL, half the squarer voltage (0.5917 V) and the logger voltage (0.5898 V) differ by 1.9 mV, a fixed gain error that P4 removes at calibration.



(a) v_1, buffer output (200 mV/div) (b) v_r, rectifier output (200 mV/div) (c) v_{sl}, log-shaped squarer output, avg −1.024 V (200 mV/div)

(d) v_{fl}, feedback logger, avg −0.519 V (100 mV/div) (e) v_l, averager output, 70.78 mV (100 mV/div) (f) V_{out}, DC output, 70.87 mV (100 mV/div)

Figure 5.2: Block outputs for case 3 (200 mVpp, 1 kHz sine): (a) inverted buffer output v_1, (b) full-wave rectified signal v_r, (c) log-shaped squarer output v_{sl}, (d) near-constant logger voltage v_{fl}, (e) averaged output v_l and (f) final DC output V_{out}. C1 (orange) shows the input and C2 (blue) as block output. All panels are at 500 µs/div.

6. Results and Discussion

The data collected from the physically implemented circuit are summarised in **Table 6.1**. The converter was tested against the calibration case and twelve test cases, with the AD3 waveform generator as the source and the output read on the AD3 oscilloscope. The cases test every input condition in **Table 3.1** at or near its limit: all seven waveform types listed in **Table 2.1** of the project specification [3], 100 and 200 mVpp, offsets from −3 V to +3 V, DC to 20 kHz and crest factors from 1 to 3. Expected values for zero-mean waveforms with an offset

follow from (6.1), with single-waveform values in **Table A.3**, and the error is defined by (6.2). The capture for each case is in **Appendix A**.

$$V_{rms,exp} = \sqrt{V_{DC}^{\ 2} + V_{AC,rms}^{\ 2}} \tag{6.1}$$

$$\%Error = \frac{V_{rms,meas} - V_{rms,exp}}{V_{rms,exp}} \times 100\% \tag{6.2}$$

Table 6.1: RMS Measurement Results

Test Case #	Waveform Type and Shape	V_{pp} / Parameter	DC Offset	Frequency	Measured RMS Value ($V_{rms,\ meas}$)	Theoretical Expected /Simulated ($V_{rms,\ exp}$)	% Error	Comment (inc. Figure reference in Appendix A)
CAL	Sine	200 mV_{pp}	+1 V	1 kHz	1.0026	1.0025	+0.01	Calibration. Fig. A.1
1	DC	—	+2 V	DC	1.998	2.000	−0.10	+DC. Fig. A.2
2	DC	—	−3 V	DC	2.983	3.000	−0.57	−3 V limit. Fig. A.3
3	Sine	200 mV_{pp}	0 V	1 kHz	0.07028	0.07071	−0.61	Max. V_{pp}. Fig. A.4
4	Sine	200 mV_{pp}	0 V	10 kHz	0.07012	0.07071	−0.84	10 kHz. Fig. A.5
5	Sine	100 mV_{pp}	0 V	20 kHz	0.03526	0.03536	−0.27	20 kHz limit. Fig. A.6
6	Triangle	100 mV_{pp}	0 V	1 kHz	0.02861	0.02887	−0.89	Min. input. Fig. A.7
7	Square	200 mV_{pp}	0 V	1 kHz	0.09925	0.1000	−0.75	CF = 1. Fig. A.8
8	Full-wave rect. sine	0–200 mV	0 V	1 kHz	0.1402	0.1414	−0.86	CF = 1.41. Fig. A.9
9	Half-wave rect. sine	0–200 mV	0 V	1 kHz	0.1006	0.1000	+0.60	CF = 2. Fig. A.10
10	PWM, 11.1 % duty	200 mV_{pp} (0–200 mV)	+100 mV	1 kHz	0.06652	0.06667	−0.22	CF = 3 limit. Fig. A.11
11	Sine	200 mV_{pp}	−2 V	5 kHz	2.003	2.0013	+0.09	−2 V + AC. Fig. A.12
12	Square	100 mV_{pp}	+3 V	10 kHz	3.007	3.0004	+0.22	+3 V limit. Fig. A.13

6.1 Accuracy Against Specifications

Overall accuracy. All thirteen measurements lie within ±1 % of expected (mean absolute error 0.46 %, largest −0.89 % in case 6), inside both the 10 % specification and the 1 % design goal. Seven of the eight AC-only cases (3–10) read low (mean −0.48 %).

DC and offset cases. Cases with a large DC component (CAL, 1, 2, 11, 12) have errors of at most 0.57 %, since the DC term dominates the sum in Eq. (6.1). Measured against its actual input of −2.948 V, the −3 V case reads 1.19 % high, which may point to a small nonlinearity in the rectifier's negative-input path at large signals.

Generator accuracy. The AD3 generator is not an exact reference: its zero-offset AC outputs carried a 2.3–2.8 mV DC offset and the case 3 input measured 70.40 mVrms against 70.71 mV nominal, 0.44 % low (Table A.1, Appendix A). Source error is therefore as large as the converter's own error. This likely explains the opposite-sign errors of the full- and half-wave rectified sines (cases 8 and 9: −0.86 % and +0.60 %).

Crest factor. No trend with crest factor is visible. The CF = 3 PWM signal (case 10) reads −0.22 %, better than the CF = 1 square wave (case 7, −0.75 %). The log-antilog core therefore tracks peaks of three times the RMS value without saturating, which is the main advantage of the implicit topology: it computes V_{in}^2/V_{out} rather than V_{in}^2, so its internal signals span the same dynamic range as the input instead of its square [1].

Frequency response. At 200 mVpp the error grows from −0.61 % at 1 kHz (case 3) to −0.84 % at 10 kHz (case 4). This is consistent with the finite slew rate and gain-bandwidth of the rectifier op-amps: at every zero crossing the op-amp output must swing across two diode drops, and that transition occupies a larger fraction of the period

as frequency rises. The 20 kHz case (case 5) reads −0.27 %. The converter nevertheless stays within 1 % up to 20 kHz.
Smallest input. The 100 mVpp triangle (case 6, 28.87 mVrms) has an error of −0.89 %, only −0.26 mV; at this level residual op-amp offset after trimming P6 and P1 is the dominant error source.

6.2 Settling Time

The settling capture (***Figure A.14, Appendix A***) shows the output after the input was switched on. From −0.40 V with no input, the output reaches 25.0 mV, within about 5 mV of its final value of about 30 mV, 0.42 s after the input is applied and shows no further visible change over the remaining 4 s. At 100 mV/div this capture cannot resolve the ±1 % band (0.3 mV), but the approach is close to exponential with a time constant of about 95 ms, which puts settling to ±1 % at roughly 0.7 s. Either way the converter settles well inside the 3 s rubric top band and the 10 s specification.

Settling is set by the averaging filter time constant, a compromise between response time and output ripple at the lowest test frequency. In log-domain RMS converters settling time increases at low signal levels [5]; this test used one of the smallest inputs (about 28 mVrms on C1), so it is close to the worst case.

6.3 Comparison with Simulation

The LTspice circuit of Section 4.3 was simulated for the same cases using fixed component values and no trimmers. Hardware and simulation agree within 0.92 % in every simulated case, with a mean absolute difference of 0.33 % (***Table 4.3.2*** against ***Table 6.1***). The largest differences occur for the full-wave rectified sine (case 8, −0.92 %), the 1 kHz sine (case 3, −0.73 %) and the smallest input (case 6, −0.66 %), and the hardware reads at or below the simulation in 10 of the 12 cases. Three effects account for this. First, the generic 2N3904 model's saturation current and emission coefficient differ from those of the real, unmatched transistors. Second, the simulation uses an ideal source, whereas the AD3 generator has its own amplitude and offset error, particularly on custom waveforms such as case 8. Third, real op-amp offsets, which the simulation omits, matter most at the smallest inputs. At 10 kHz, by contrast, the two agree within 0.02 % (case 4: −0.84 % measured, −0.82 % simulated). This confirms that the high-frequency error is a property of the rectifier design rather than a measurement artefact.

6.4 Challenges and Limitations

Interacting trimmer potentiometers. Calibration used four multi-turn trimmer potentiometers whose adjustments affect each other: P6 shifts positive and negative readings in opposite directions, P1 shifts all readings equally, P5 sets the negative-input gain and P4 sets the overall gain. Trimming them in the order P6, P5, P4, P1 (buffer offset, negative-input gain, overall gain, then output offset) avoided repeated iterations, and the final settings are recorded in ***Table A.2, Appendix A*** so the calibration can be restored if the board is disturbed.
Thermal drift. Although nV_T cancels around the loop, the discrete transistors are side by side but not thermally bonded. A 1 °C difference between them shifts V_{BE} by about 2 mV, the same size as the 1.9 mV mismatch measured at CAL in ***Section 5.2 (Sub-Block Verification)***, producing a gain error of a few percent. Placing the translinear transistors in thermal contact, or using a matched pair, would reduce the drift.
Source accuracy. The AD3's amplitude and offset errors are comparable to the converter's own error, so the input measured on C1, rather than the generator setting, should be the reference for every case.

7. Conclusion

A fully analogue true RMS-to-DC converter was designed, simulated, and built using only kit components and the ±5 V supplies of the Analog Discovery 3. Following the literature survey, an implicit log–antilog architecture was selected because it handles the required 100:1 input range that an explicit squarer cannot. The design combines a translinear squarer/divider, similar to that of the AD536A, with AD637-style op-amp averaging, implemented in six blocks from discrete 2N3904 transistors and TL084/TL974 op-amps.

The hardware met the project's accuracy, input-range, frequency, and settling specifications, as well as the stricter design goals. After a single calibration, all thirteen test cases were within ±1% of their expected values, with a mean absolute error of 0.46%. This included the smallest 28.9 mV input (−0.89%), a crest factor of 3

(−0.22%), DC offsets of ±3 V, and frequencies up to 20 kHz. The output settled well within 1 s, comfortably inside the 3 s design goal, and hardware results agreed with simulation to within 0.92%.

The main limitations stem from building with discrete parts. The translinear transistors were unmatched and not thermally bonded, so their mismatch and drift were compensated by calibration rather than eliminated. Four interacting trimmers were required, and residual op-amp offset dominated the error at the smallest inputs. Accuracy also fell slightly at higher frequencies, consistent with rectifier slewing at zero crossings, and settling was measured only for an increasing input step.

Future work should use a matched transistor array or thermally bonded devices, lower-offset op-amps to reduce reliance on trimming, and a faster rectifier op-amp. Testing should also include decreasing input steps and frequency sweeps at a constant amplitude, to characterise the converter's limits more completely.

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APPENDIX A

This appendix contains the Waveforms capture for each case in Table 6.1 (Figs. A.1–A.13), the settling-time capture (Fig. A.14), the block-level measurements and calibration record referred to in Section 6, and the expected-value calculations. In every capture C1 (orange) is the converter input and C2 (blue) the DC output, and the cursor readouts give the values in Table 6.1. Scales are 500 µs/div and 50 mV/div unless the caption states otherwise.

A.1 Test-Case and Settling Captures

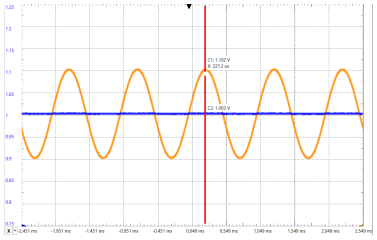


Figure A.1: CAL: sine, 200 mV_{pp}, +1 V, 1 kHz

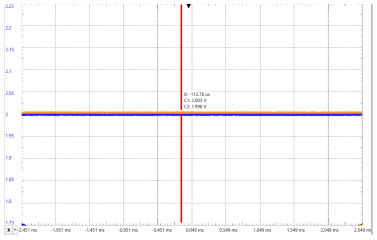


Figure A.2: Case 1: DC, +2 V

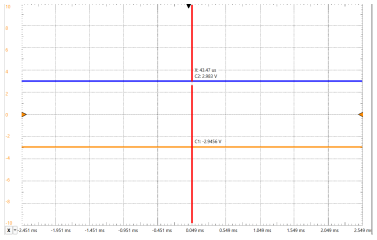


Figure A.3: Case 2: DC, −3 V (2 V/div)

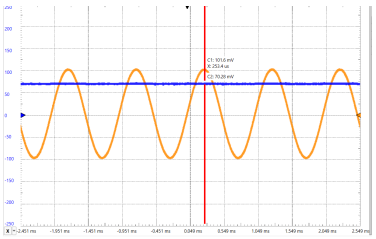


Figure A.4: Case 3: sine, 200 mV_{pp}, 1 kHz

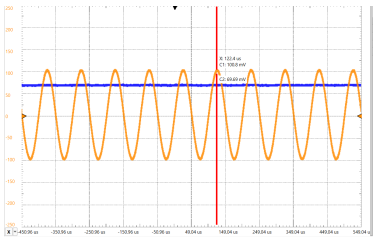


Figure A.5: Case 4: sine, 200 mV_{pp}, 10 kHz (100 µs/div)

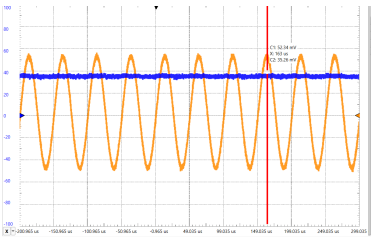


Figure A.6: Case 5: sine, 100 mV_{pp}, 20 kHz (50 µs/div, 20 mV/div)

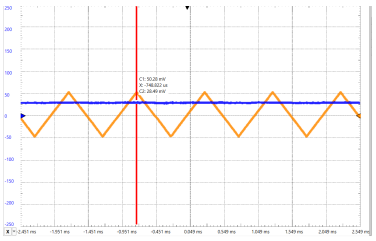


Figure A.7: Case 6: triangle, 100 mV_{pp}, 1 kHz

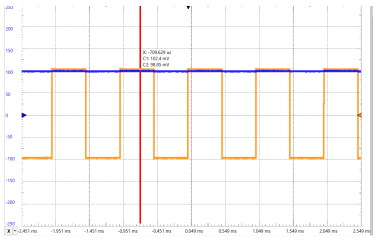


Figure A.8: Case 7: square, 200 mV_{pp}, 1 kHz

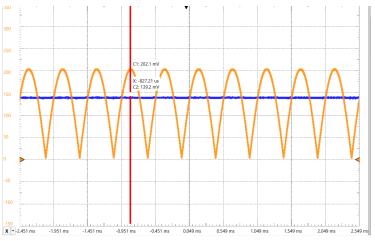


Figure A.9: Case 8: full-wave rectified sine, 1 kHz

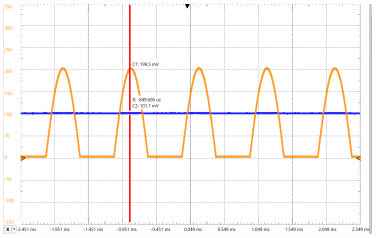


Figure A.10: Case 9: half-wave rectified sine, 1 kHz

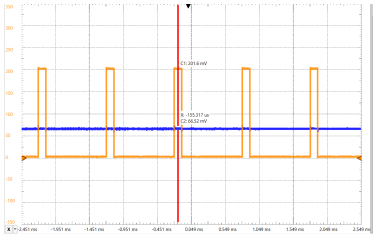


Figure A.11: Case 10: PWM 11.1 %, 0–200 mV, 1 kHz

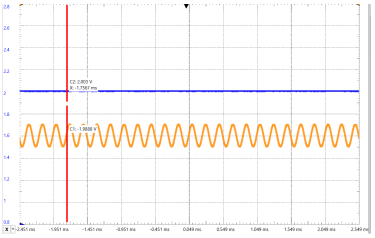


Figure A.12: Case 11: sine, 200 mV_{pp}, −2 V, 5 kHz (200 mV/div; C1 has its own display offset)

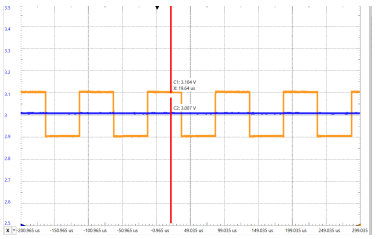


Figure A.13: Case 12: square, 100 mV_{pp}, +3 V, 10 kHz (50 µs/div, 100 mV/div)

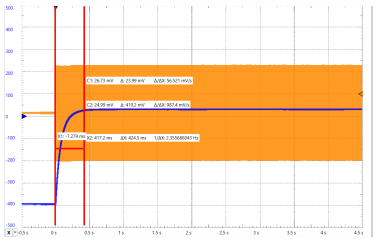


Figure A.14: Settling: input switched on at t = 0; C2 rises from −0.40 V (500 ms/div, C2 100 mV/div)

A.2 Block-Level Measurements

Table A.1: Block outputs for the calibration case and case 3

Node	Block	CAL expected	CAL measured	Case 3 expected	Case 3 measured	Quantity (case 3)
V_{in}	Input	1.0025 V	1.0031 V	70.71 mV	70.40 mV	RMS
v_l	0 – Buffer	−1.000 V	−1.0054 V	70.71 mV, inverted	70.25 mV	RMS
v_r	1 – Rectifier	1.000 V	0.9985 V	63.66 mV ($2/\pi \times$ 100 mV)	60.47 mV	Average
v_{sl}	2 – Squarer	−1.1 to −1.3 V ($-2V_{BE}$)	−1.1834 V	Log-shaped waveform	Log shape as expected	Waveform
v_{fl}	3 – Feedback logger	−0.55 to −0.65 V ($-V_{BE}$)	−0.5898 V	≈ constant ($-V_{BE}$)	−0.5189 V	Average

v_1	4 – Averager	1.0025 V	1.0031 V	70.71 mV, small ripple	70.78 mV	Average
V_{out}	5 – Output	1.0025 V	1.0038 V	70.71 mV	70.87 mV	Average

Note: these block-level readings were taken in a separate measurement from Table 6.1, so the output values differ slightly.

A.3 Calibration Record

Table A.2: Final trimmer settings

Trimmer	Block	Function	Final setting
P6	0	Buffer offset before the rectifier; shifts positive and negative readings in opposite directions	99.8 kΩ
P5	1	Rectifier gain for negative inputs	0.82 kΩ
P4	3	Overall gain	0.65 kΩ
P1	2	Offset after the rectifier; shifts all readings by the same amount	54.8 kΩ

A.4 Expected-Value Calculations

Table A.3: RMS of the test waveforms (V_p = peak amplitude)

Waveform	V_{rms}	Test cases	Example
DC	V_{DC}	1, 2	2 V → 2.000 V
Sine (zero mean)	$\frac{V_p}{\sqrt{2}}$	3, 4, 5 (CAL, 11 via 6.1)	V_p = 100 mV → 70.71 mV
Triangle (zero mean)	$\frac{V_p}{\sqrt{3}}$	6	V_p = 50 mV → 28.87 mV
Square (zero mean)	V_p	7 (also 12, with DC offset added via Eq. 6.1)	V_p = 100 mV → 100.0 mV
Full-wave rectified sine	$\frac{V_p}{\sqrt{2}}$	8	V_p = 200 mV → 141.4 mV
Half-wave rectified sine	$\frac{V_p}{2}$	9	V_p = 200 mV → 100.0 mV
PWM, duty t_1/T	$V_p\sqrt{t_1/T}$	10	V_p = 200 mV, t_1/T = 1/9 → 66.67 mV (CF = 3)

For the zero-mean waveforms with an added DC offset (CAL, 11, 12), Eq. (6.1) in Section 6, Vrms = $\sqrt{(V_{DC}^2 + V_{AC,rms}^2)}$, combines the two components; for example, case 12 gives $\sqrt{(3^2 + 0.05^2)}$ = 3.0004 V. The rectified sines and the PWM signal (cases 8–10) are unipolar and their Vrms already include the DC component, so Eq. (6.1) is not applied again.

APPENDIX B

Team Contribution Matrix and Weekly Progress Minutes

Team ID	Team Members Name (Initial)	Student ID #	Significant Individual Contribution (Describe the main sections you worked on during the project)
Group A2	Wu Li Min (LiMin)	34478841	Circuit design, build, testing and debugging (Blocks 0–5); data recording; Report Sections 5, 6 and References
Group A2	Nicholas Wong Zhong Hao (Nicholas)	35559004	Circuit design, build, testing and debugging (Blocks 0–5); data recording; Report Sections 5, 6 and References
Group A2	Yeoh Jie Ying (JieYing)	34601643	Circuit design, LTspice simulation and testing (Blocks 0–5); Report Section 4 (Design Description)
Group A2	Joyce Ngaa Chee Siew (Joyce)	34600183	Circuit design, LTspice simulation and testing (Blocks 0–5); Report Section 3 (Design Specifications)
Group A2	Hui Man Kitt (Mankitt)	34475044	Report Sections 1 (Introduction), 2 (Literature Survey) and 7 (Conclusion)

Weekly Progress & Meeting Minutes (one row per week, 2026; all meetings 8:30–11:30 am, all members present)

Week #	Meeting date	Attendees Initials	Agenda / discussion points	Key decisions (what/why)	Actions & owners (Task IDs – tracking)	Issues raised / Previous meeting follow-up	Resolutions / next steps
7	10/09	All	Spec review; RMS methods survey	Implicit log–antilog topology: wide dynamic range	Survey – Mankitt; LTspice – JieYing, Joyce; parts – LiMin, Nicholas	Initial circuit complex; simulation off target	Research simpler circuits [1]; re-simulate
8	17/09	All	Review circuit; map to kit parts	2N3904 as diodes (1N4007 too slow); TL084 for logs, TL974 for rectifier/output	LTspice – JieYing, Joyce; build – LiMin, Nicholas	W7 follow-up: circuit simulated	Full build in W9
9	24/09	All	Build and test Blocks 0–5	Test each block before closing the loop	Build/test – LiMin, Nicholas, JieYing, Joyce	3 TL974 damaged (supply > 12 V or short); errors +53 % / −8 %	ICs replaced; ±5 V checked; errors traced to offset/gain
10	08/10	All	Calibration; test cases	Trim order P6 → P5 → P4 → P1; CH1 as reference	Tests – LiMin, Nicholas; sim. comparison – JieYing, Joyce	W9 follow-up: errors trimmed, CAL 1.0026 V	13 cases within ±1 %; draft 11/10
11	15/10	All	Review draft vs template	Remove template text; renumber; IEEE references	Sec. 6.3 – JieYing, Joyce; Sec. 7 – Mankitt	Over page limit; Sec. 6.3, 7 missing	Condensed; sections completed
12	20/10	All	Rehearsal; final recalibration	Each member presents own work	Rehearsal – All	W11 follow-up: report complete	Present 22/10; submit 23/10; return kits

APPENDIX C

Generative AI usage. Tool & model for all uses: Claude Sonnet 5.5 (Anthropic). All outputs were reviewed and verified by the team; Use IDs C1–C3 are cited in the report text.

Use ID	Purpose / project stage	Where in report	Output type	How used	Verification (brief)	Sources	Initials & date	Example prompt (translated)
C1	Literature survey: summary of the ADI RMS-to-DC guide	Sec. 2.1–2.4	Text	Guidance only	Guide read; equations re-derived	[1]	Mankitt, 29/09	"Explain the RMS-to-DC guide and which part can be used for the project."
C2	Test planning: test cases, expected RMS, AD3 setup, debugging	Table 6.1; Table A.3	Tables, values	Guidance only	Recomputed by hand; verified on AD3	[3]; AD3	LiMin, Nicholas, 29/09	"Generate expected results for each waveform at 1 kHz, 200 mVpp and 100 mVpp."
C3	Report review: wording, consistency, IEEE references	Sec. 2–7; References	Text	Guidance only	Checked against [1], datasheets and data	[2]–[8]	All, 11/10	"Check my report against the formal format and project requirements."

Bill of Materials

All parts come from the ECE3161 unit kit. Designators follow **Figure 5.1** (U, T, P) and the circuit schematic in Section 4 (R, C).

Table BOM.1: Bill of materials

#	Part	Description	Designators	Qty
1	TL084CN	Quad JFET-input op-amp	U1 (Block 0), U3 (Blocks 2–3)	2
2	TL974IN	Quad low-noise op-amp	U2 (Block 1), U4 (Blocks 4–5)	2
3	2N3904	NPN BJT: translinear log/antilog devices and diode-connected rectifier and clamp diodes	T1–T8	8
4	1 MΩ, 1 % metal film	Buffer input and feedback; offset-injection resistors	R_{in1}, R_{f1}, R_{T1}, R_{T2}	4
5	100 kΩ, 1 % metal film	Paralleled with 10 kΩ to trim gain-setting values	R_{2b}, R_5	2
6	10 kΩ, 1 % metal film	Rectifier gain network (20 kΩ = 2 × 10 kΩ), squarer, logger, averager and output filter	R_{2a}, R_{2b}, R_{3b}, R_{f3} (×2), R_{3a} (×2), R_{sl}, R_5, R_7, R_p	11
7	1 kΩ, 1 % metal film	Log-amp emitter resistors	R_{E4}, R_{E5}	2
8	100 kΩ multi-turn trimmer (W104)	Offset trims	P1, P6	2
9	1 kΩ multi-turn trimmer (W102)	Rectifier balance and gain calibration	P5, P4	2
10	10 µF electrolytic	Averaging and output filter; bulk rail bypass	C_7, C_p, C_{b1}, C_{b2}	4
11	100 nF ceramic	Supply decoupling, one per supply pin	2 per IC	8
12	22 pF ceramic	Log-amp compensation	C_{c4}, C_{c5}	2
13	Breadboard	400-point (30-row) solderless breadboard	—	2
14	Solid-core jumper wire	22–26 AWG hook-up wire	—	As required

End of Report